

MPT Data-Driven Asset Allocation Model

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Project Methodology

A 5-year database containing 77,338 datapoints (from December 2020 to December 2025) was gathered via the CHASS/TSX database system containing daily closing prices for the TSX 60.

An initial python script was developed using the python pandas library to load the exported CSV data into computer memory and clean the data (i.e. converting written dates into datetime data, removing NAN data).

A script was written using the scipy library's built-in minimization optimizer toolset to run an SLSQP optimization algorithm to calculate the Sharpe optimized weights for the TSX 60 using a risk free rate of 2.68% (based on the Canadian 1-Year Treasury). Additionally, the script ran another algorithm to calculate the risk-minimizing weights.

Using yfinance's public API, data was pulled on two funds and on the closing prices of the two optimized funds between the out-of-sample period from January 1, 2026 to May 24, 2026:

- iShares S&P/TSX 60 Index ETF (XIU.TO): The TSX 60 proportionally by float-adjusted market cap.
- TSX Composite (^GSPTE): 220 publicly listed stocks on the TSX based on float-adjusted market cap.

Using matplotlib, the outcome can be visualized. The 4 year training dataset was restricted to 4, 3, 2, 1, and ½ years to test the hypothesis that recency is important to the accuracy of the model. (See Figs.4-9 below)

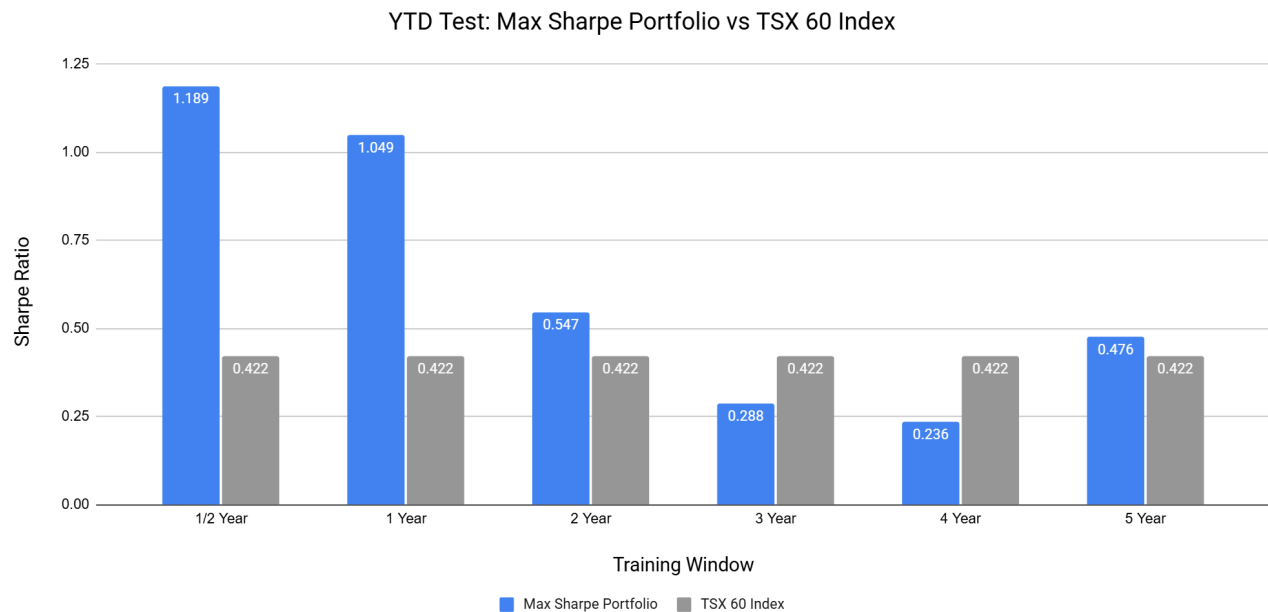


Fig.1 Sharpe Ratio Performance of Max Sharpe Portfolio vs TSX 60 by Training Window Size

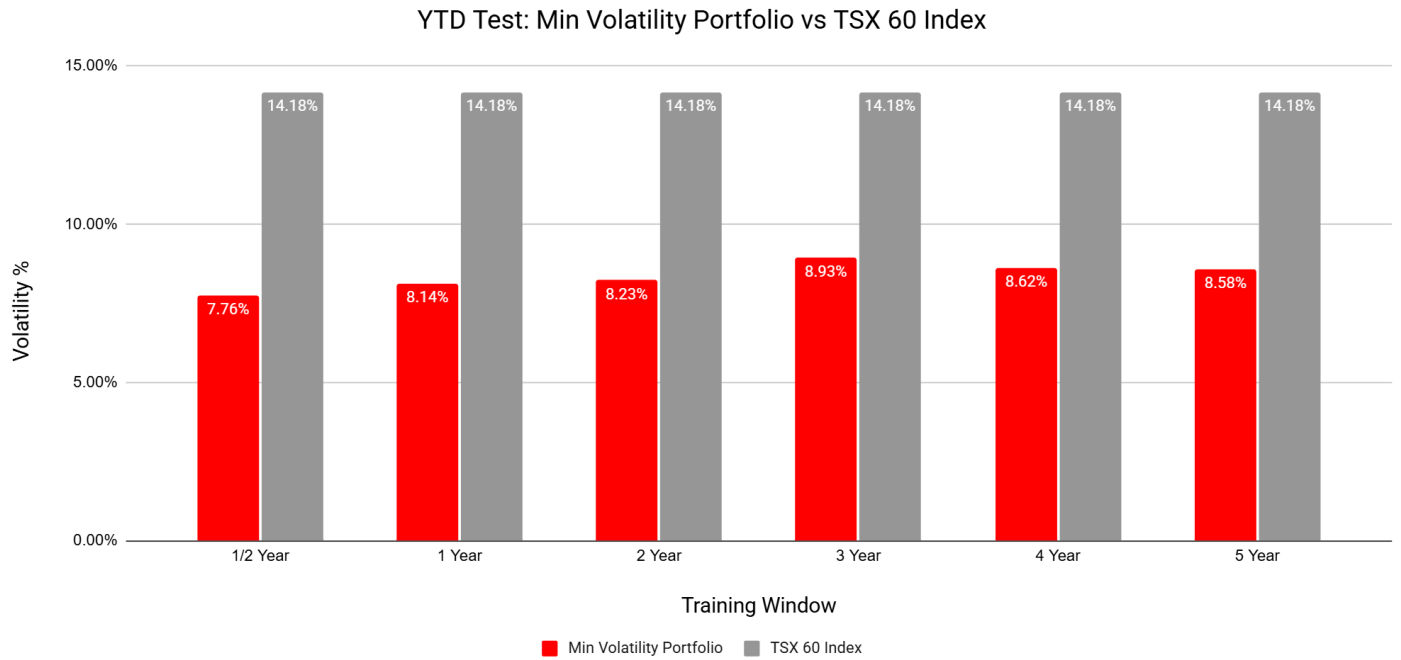


Fig.2 Volatility Performance of Min Volatility Portfolio vs TSX 60 by Training Window Size

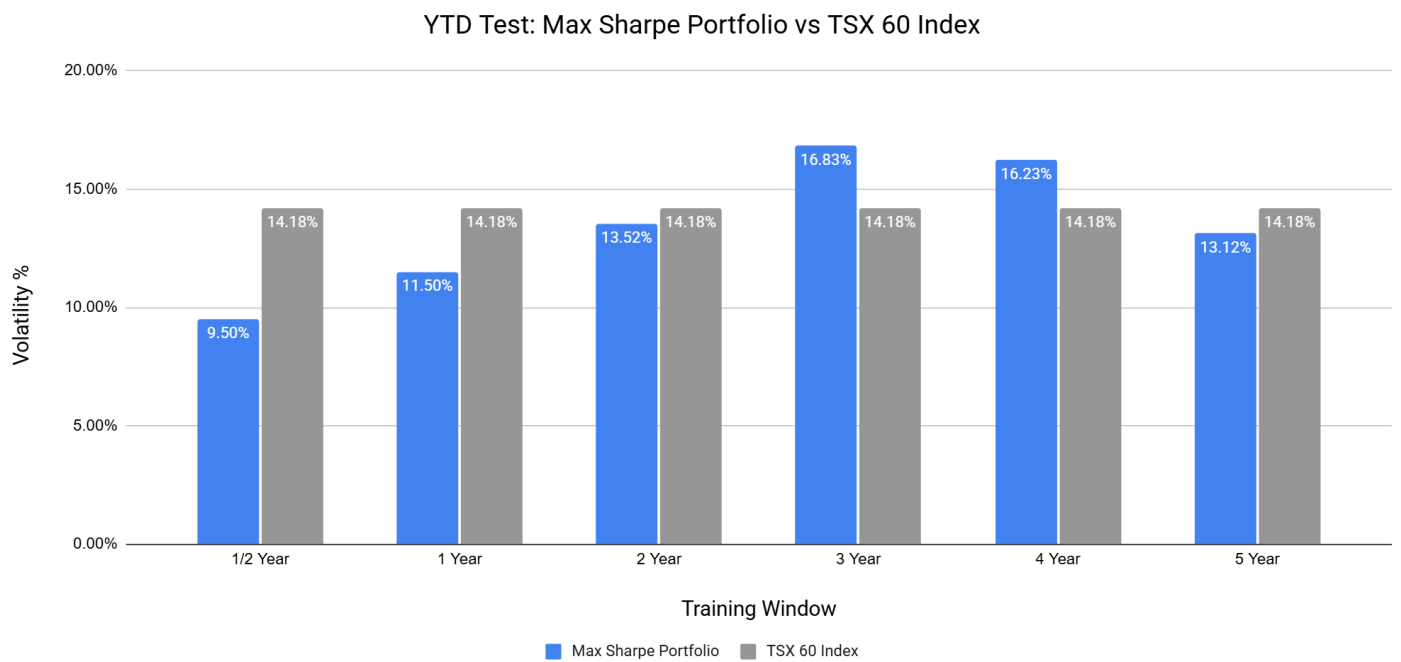


Fig.3 Volatility Performance of Max Sharpe Portfolio vs TSX 60 by Training Window Size

Hypothesis Testing

To properly assess the validity of our findings (our two best, ½ year training and 1 year training) we need to run a proper Jobson-Korkie test (hypothesis test designed to validate Sharpe ratios). We'll compare the ½ year Max Sharpe optimized portfolio with the iShares TSX 60 Index weighted by market cap since they both occupy the same universe of investments. Let's start by calculating the variance:

(Note. iShares TSX 60 and Max Sharpe portfolios produced the following data)

$$SR_a = 1.189, SR_b = 0.422, r = 0.7404, n = 126$$

$$Var(SR_a - SR_b) = \frac{1}{n} \left(2 - 2r + \frac{SR_a^2}{2} + \frac{SR_b^2}{2} - SR_a^2 \cdot SR_b^2 \cdot r^2 \right)$$

$$Var(SR_a - SR_b) = \frac{1}{126} \left(2 - 2(0.7404) + \frac{1.189^2}{2} + \frac{0.422^2}{2} - 1.189 \cdot 0.422 \cdot 0.7404^2 \right)$$

Calculate the standard error based on the variance formula above:

$$SE = \sqrt{Var(SR_a - SR_b)}$$

$$SE = 0.09085$$

Calculate the Z-Statistic of the test:

$$z = \frac{1.189 - 0.422}{0.09085}$$

$$z = 8.44$$

And the associated probability:

$$p \approx 0.0000000000000003$$

A Jobson-Korkie test applied to the 126-day out-of-sample return series yields a z-statistic of 8.44 ($p \approx 0$), suggesting statistically significant outperformance over this test window. Two limitations apply:

- First, the optimization implicitly fits historical patterns that may not persist (overfitting risk / equilibrium assumption).
- Second, a single 126-day window is borderline for the normality assumptions underlying the test, and insufficient to distinguish a genuine edge from a favorable market regime.

Replicating across multiple non-overlapping out-of-sample periods spanning different market conditions would substantially strengthen this conclusion.

Reflection and Further Improvement

This project is simplistic, uses assumptions that historical data is equivalent to future data (called the equilibrium assumption) and is trained on a small, limited-universe set of only public equity data. Although the results are promising, showing:

- Training **within 1 year's** worth of data shows the highest Sharpe ratios, strong evidence for a hypothesis suggesting recent price action in the TSX 60 is highly relevant compared to older
- **Most optimal** results yielded a sharpe ratio of 1.189 for a model trained on 0.5 years of price data (**13.98% YTD return, 9.50% Volatility**)
- **The TSX 60** distributed by market cap is not the most optimal distribution of weights for investments (**8.66% YTD Return, 14.18% Volatility**)
- Training on **3 or 4 years of data yields results worse** than the TSX 60 provides worse models
- Training on a 5 year window of data yields better results than the TSX 60 but worse results than the 1 and 0.5 year models

However, these results are not stable and have much more room for improvement. Here are some actionable next steps to improve the model:

1. **Increase the universe of assets** to bonds, real estate, and other investment bodies with accurate data to increase the efficiency frontier
2. Test even **larger windows of data**, 6-10 years
3. Continue gathering evidence to support the relevance hypothesis by testing variable training windows in a different timeframe (i.e. this project was done on a 2020-2026 time window, **redo everything on data from 2014-2020** and perform the same **out-of-bounds testing**)
4. Implement the **Black-Litterman model** (allowing user sentiment to adjust weights based on predictions and confidence scoring) to challenge the equilibrium assumption
5. Set up an NLP model (Such as **FinBERT**) that is given scraped data from financial reporting (SEC, CEDAR+ data, Financial News) to **automate and standardize the Black-Litterman extension**
6. Explore avenues to tackle portfolio concentration risk (i.e. increasing the universe of assets, adjusting inequality constraints to limit the weight of one single stock.

In conclusion, this project delivered acceptable results, minimizing volatility and increasing returns. However there is much room for expansion and optimization, challenging biases and bad assumptions in the current modelling can yield even greater performance.

5 Year Training Window

	YTD Return	Volatility	Sharpe Ratio (2.68% RFR)
Max Sharpe	8.93%	13.12%	0.476
Min Volatility	5.61%	8.58%	0.342
TSX 60 Index	8.66%	14.18%	0.422
TSX Composite	8.12%	15.78%	0.345

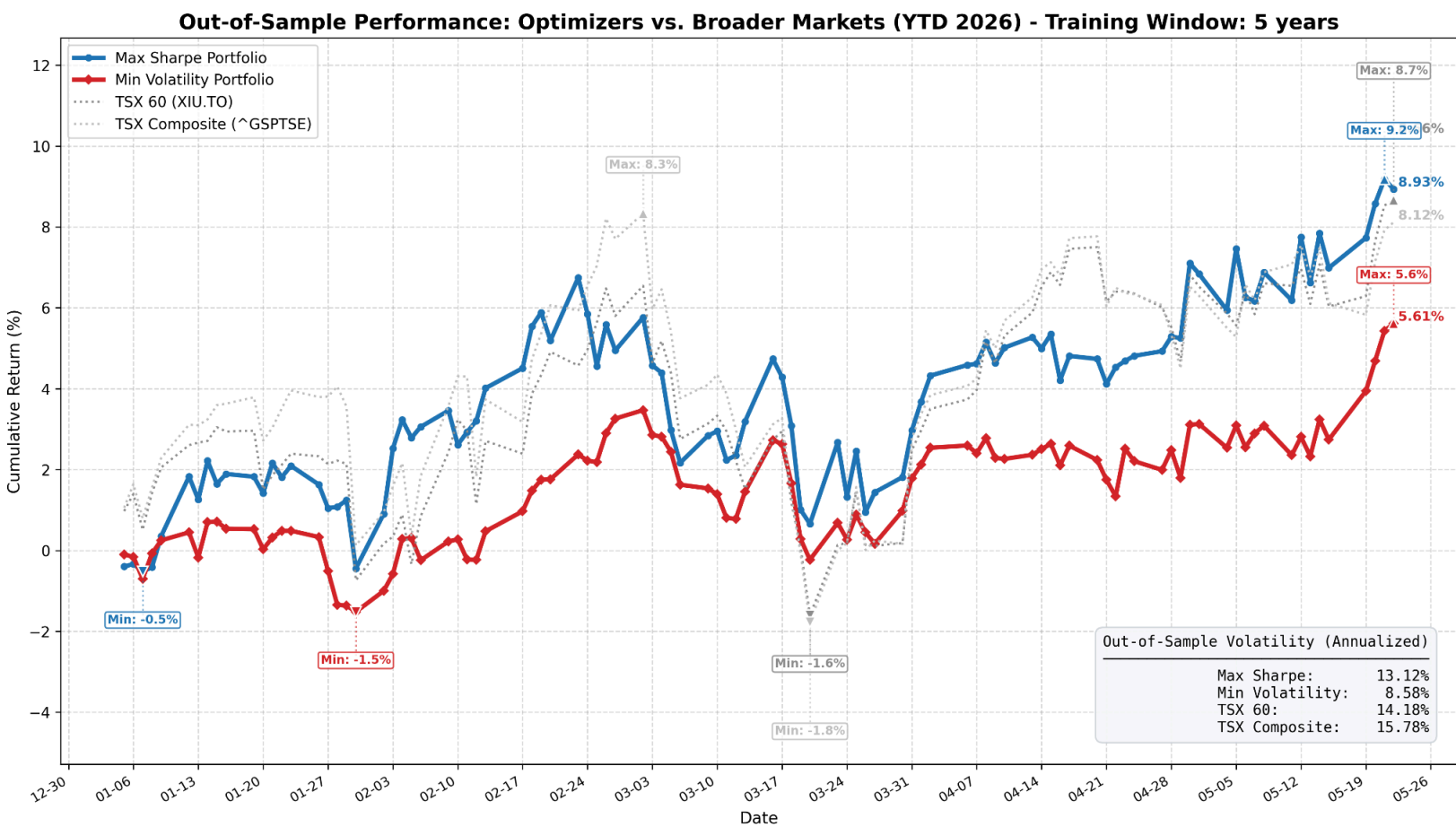


Fig.4 Performance Chart of Two Optimized Funds Against Two Comparisons. 5 Years

4 Year Training Window

	YTD Return	Volatility	Sharpe Ratio (2.68% RFR)
Max Sharpe	6.51%	16.23%	0.236
Min Volatility	5.43%	8.62%	0.319
TSX 60 Index	8.66%	14.18%	0.422
TSX Composite	8.12%	15.78%	0.345

Out-of-Sample Performance: Optimizers vs. Broader Markets (YTD 2026) - Training Window: 4 years

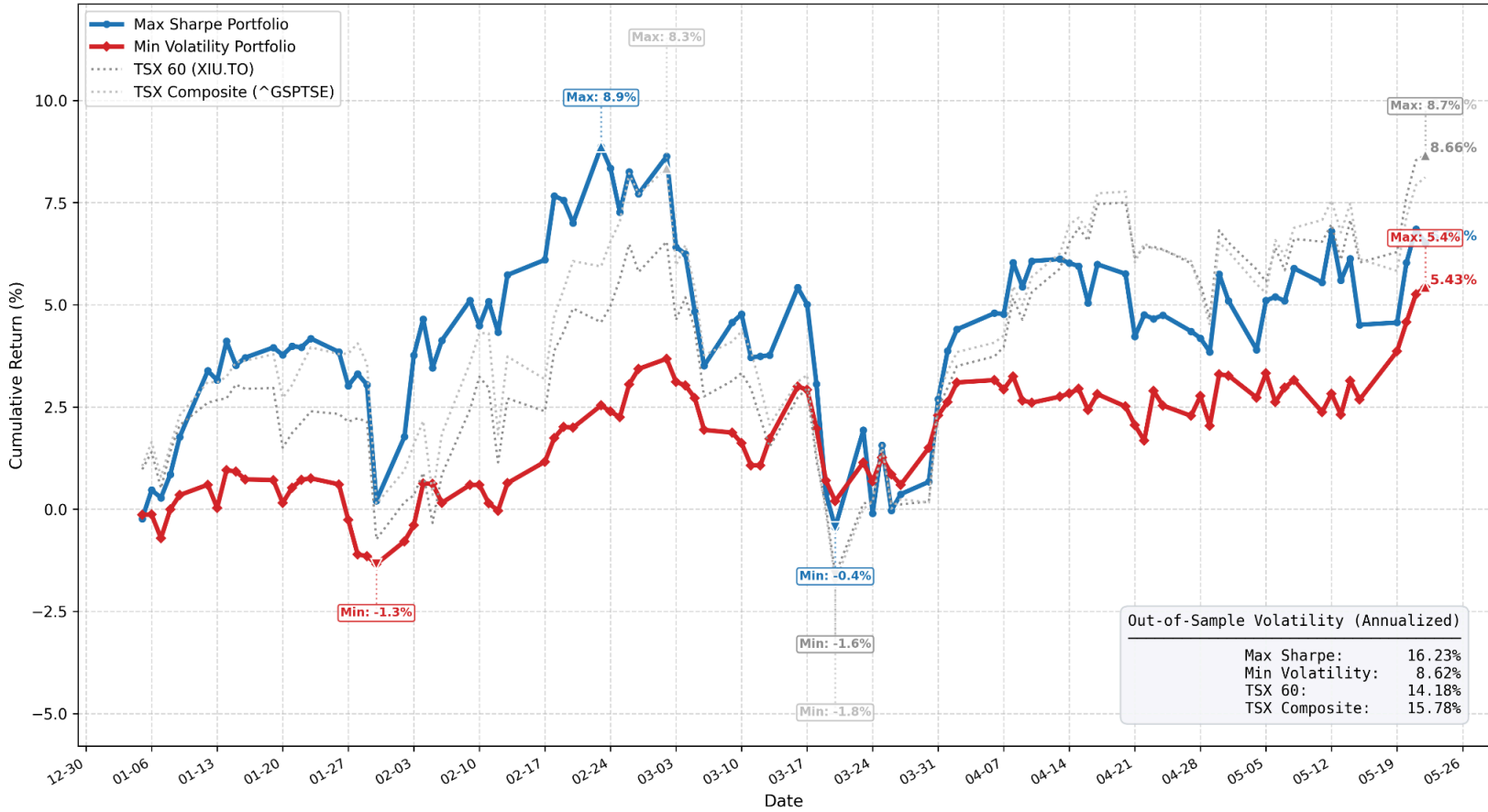


Fig.5 Performance Chart of Two Optimized Funds Against Two Comparisons. 4 Years

3 Year Training Window

	YTD Return	Volatility	Sharpe Ratio (2.68% RFR)
Max Sharpe	7.53%	16.83%	0.288
Min Volatility	5.84%	8.93%	0.354
TSX 60 Index	8.66%	14.18%	0.422
TSX Composite	8.12%	15.78%	0.345

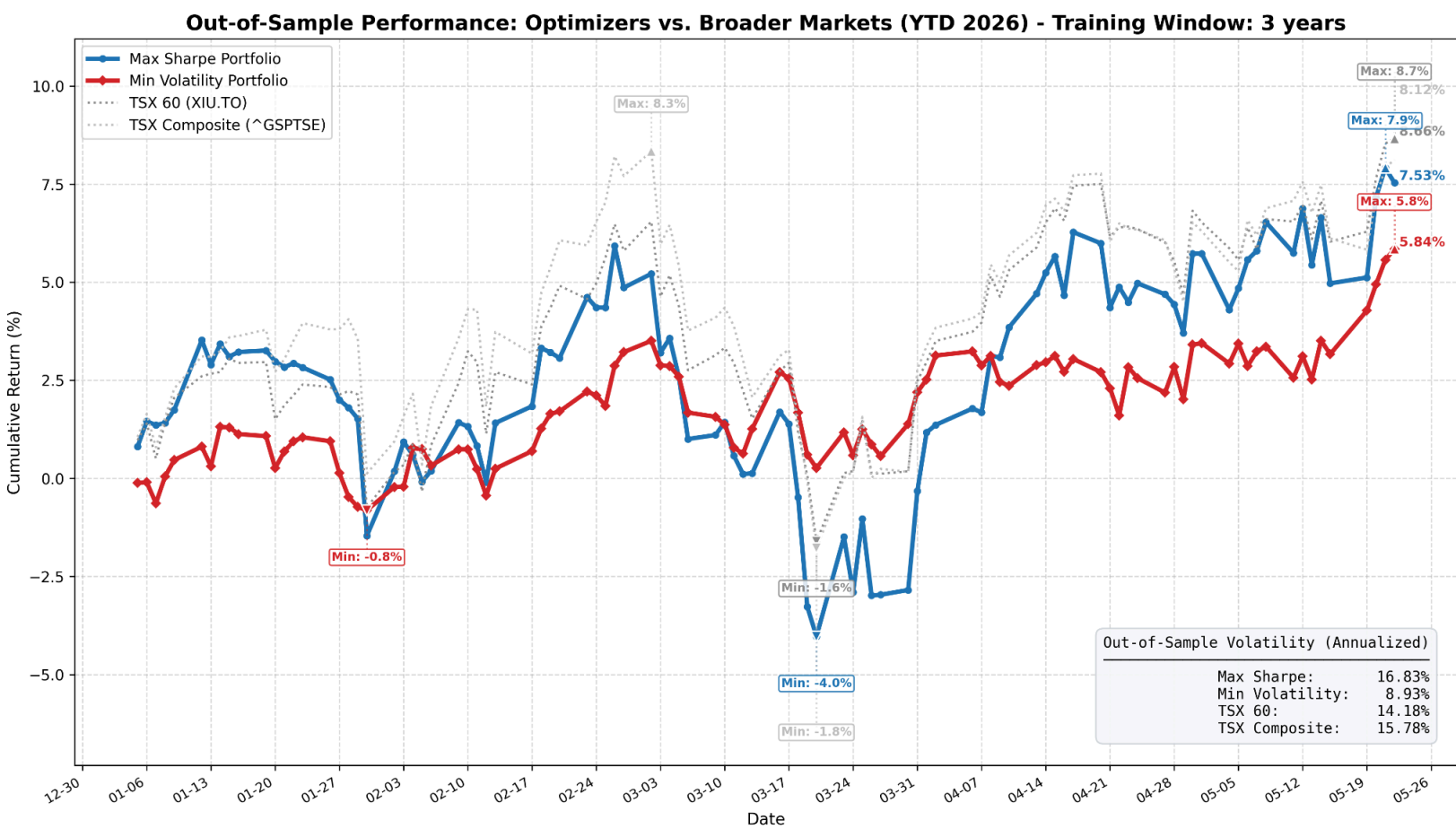


Fig.6 Performance Chart of Two Optimized Funds Against Two Comparisons. 3 Years

2 Year Training Window

	YTD Return	Volatility	Sharpe Ratio (2.68% RFR)
Max Sharpe	10.07%	13.52%	0.547
Min Volatility	7.90%	8.23%	0.634
TSX 60 Index	8.66%	14.18%	0.422
TSX Composite	8.12%	15.78%	0.345

Out-of-Sample Performance: Optimizers vs. Broader Markets (YTD 2026) - Training Window: 2 years



Fig.7 Performance Chart of Two Optimized Funds Against Two Comparisons, 2 Years

1 Year Training Window

	YTD Return	Volatility	Sharpe Ratio (2.68% RFR)
Max Sharpe	14.74%	11.50%	1.049
Min Volatility	10.14%	8.14%	0.916
TSX 60 Index	8.66%	14.18%	0.422
TSX Composite	8.12%	15.78%	0.345

Out-of-Sample Performance: Optimizers vs. Broader Markets (YTD 2026) - Training Window: 1 years

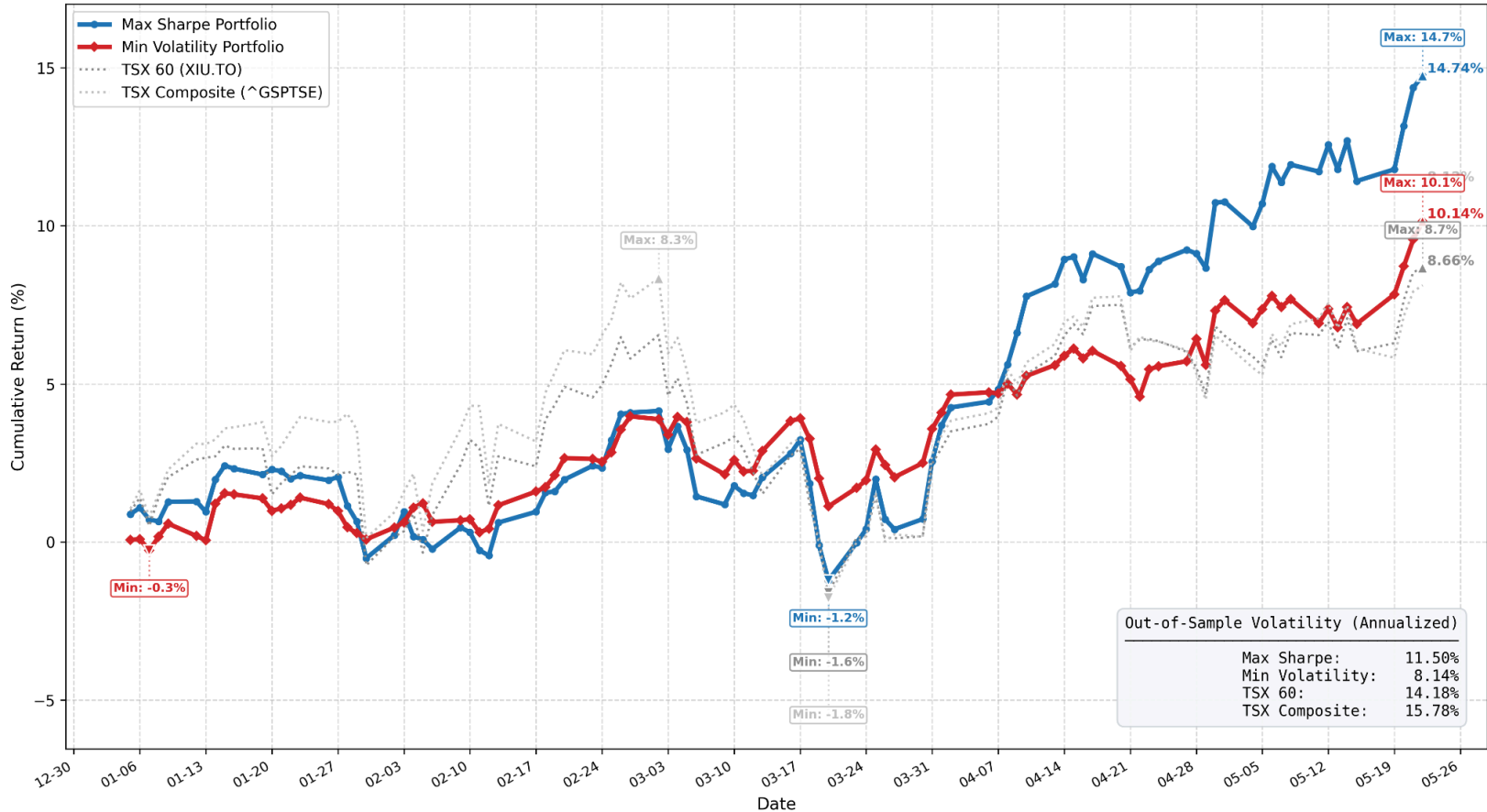


Fig.8 Performance Chart of Two Optimized Funds Against Two Comparisons, 1 Year

0.5 Year Training Window

	YTD Return	Volatility	Sharpe Ratio (2.68% RFR)
Max Sharpe	13.98%	9.50%	1.189
Min Volatility	11.44%	7.76%	1.129
TSX 60 Index	8.66%	14.18%	0.422
TSX Composite	8.12%	15.78%	0.345

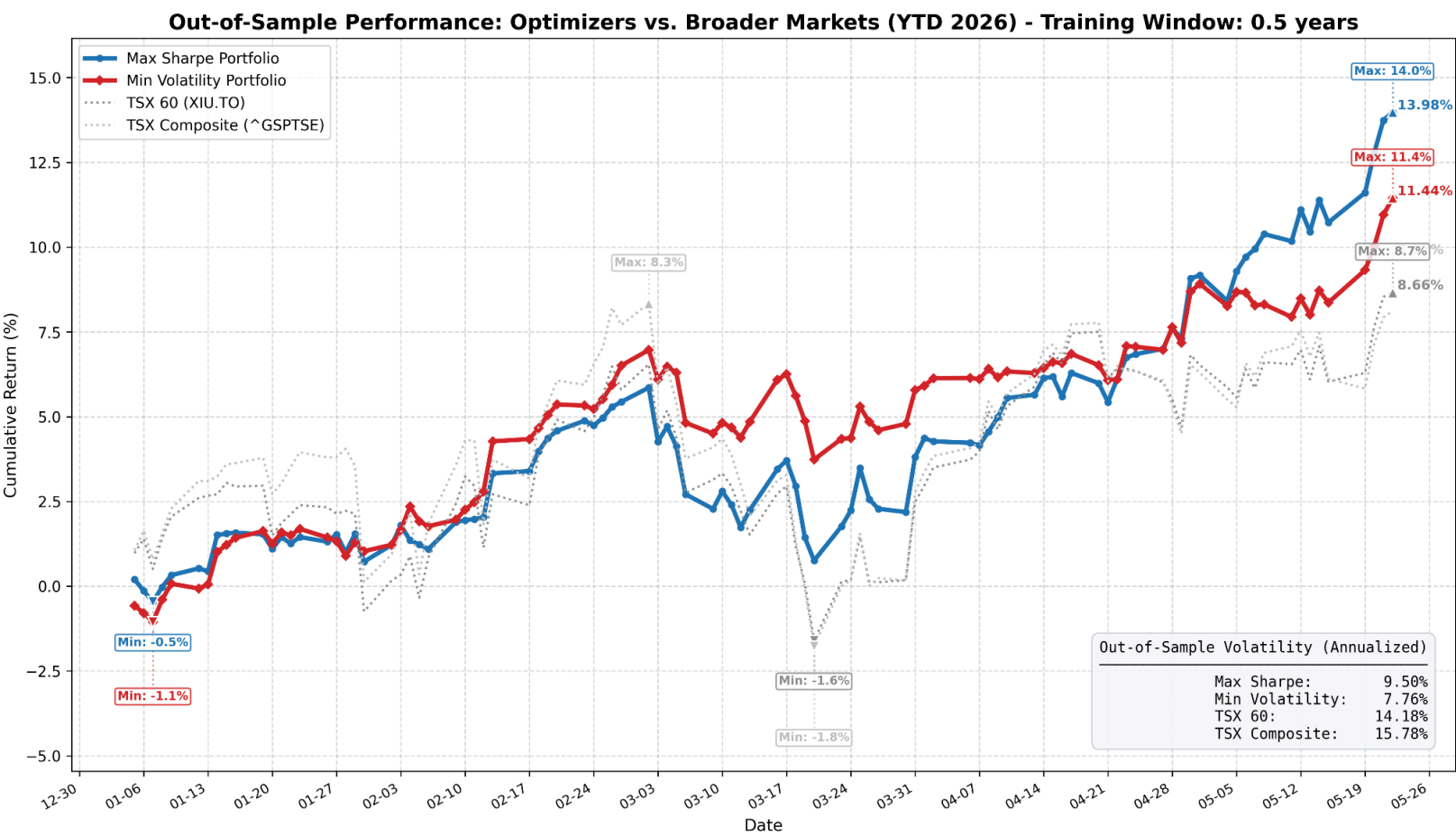


Fig.9 Performance Chart of Two Optimized Funds Against Two Comparisons, 0.5 Year

0.5 Year Window Portfolio Distribution

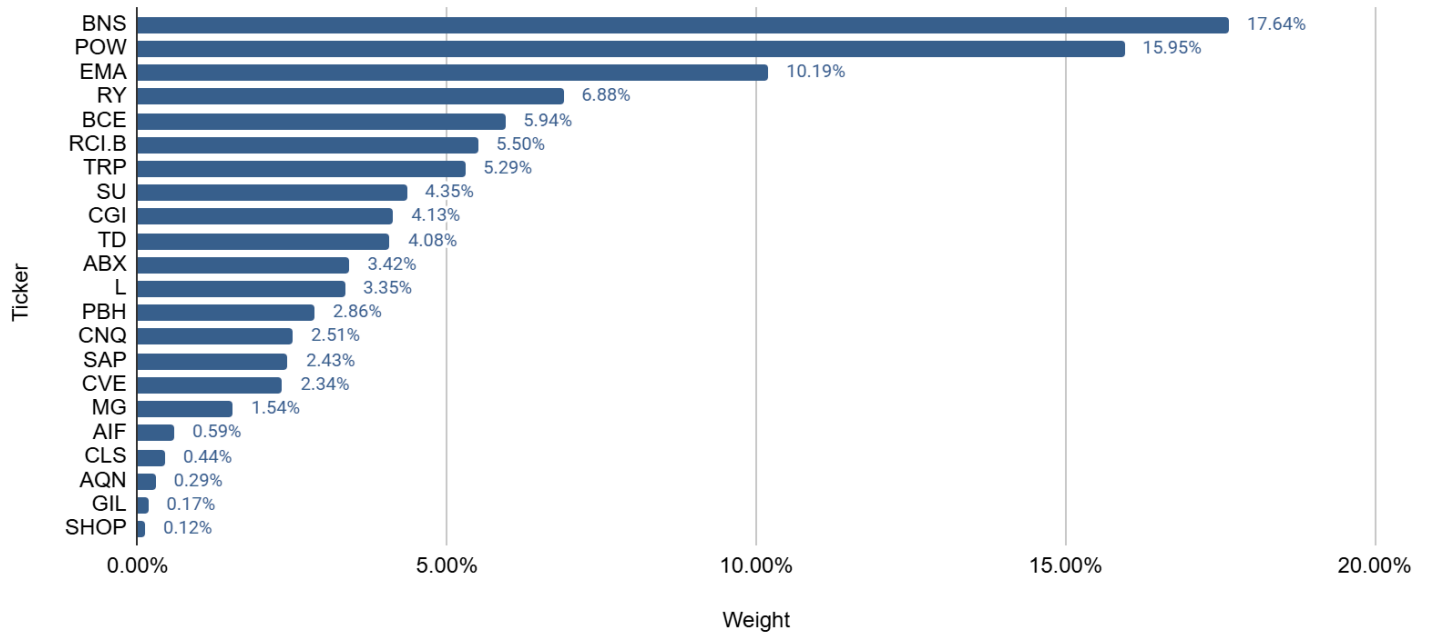


Fig.10 Percent Weight Distribution of The Max Sharpe Portfolio, 0.5 Year